

Define the Work

Step 1: Write Down What the Cost Depends On

Assumptions	Why it drives cost	Owner
Cross-platform codebase (iOS & Android)	Building from a single shared codebase avoids duplicating frontend engineering hours, directly lowering development labor costs	Technical Lead
Computer vision model processing handles real-time video feeds	Requires specific technical infrastrucutre, higher computing and optimization to process live form detection without severe latency	Technical Lead
Core API and user account back-end run on cloud infrastructure	Determines hosting, server scaling, and database storage tiers required to manage user history and stats over time.	Technical Lead/DevOps
Third-party integrations (payments)	Every added external service introduces implementation complexity, testing overhead, and potential API/transaction fees.	Integration lead
Fixed project duration and defined team capacity	Sets the total labor footprint, project milestones, and resource allocation across sprints	Project Sponsor/Manager

Step 2: Break the Work Into Packages

Work Package	Role	Hours
Real-time video processing & form detection engine	Developers	1200
User interface & workout dashboard (UX/UI)	UX Designer/Developers	700
User account management & workout history back end (APIs)	Developers	900
Testing, usability evaluation, and quality assurance (QA)	QA	600
Cloud infrastructure, video streaming, and deployment (DevOps)	Development operations	400
Total		3800

Step 3: Estimate Effort More Than One Way

$E = (O + 4M + P) \div 6$

O = optimistic: 3,200 hours

M = most likely: 3,700 hours

P = pessimistic: 4,600 hours

$E = \div 6 = 3,767$ hours

3,767

User story	MoSCoW	Points	Column 1
As a user, I can grant camera permission and start a live-feedback session.	Must	5	
As a user, I can see the webcam stream on the workout screen.	Must	5	
As a user, I can see simulated form feedback overlaid on the live video.	Must	3	
As a user, I receive a warning when feedback is delayed or disconnected, and	Should	5	
As a user, I can enable an optional audio cue when new feedback appears.	Could	3	
Total points		23	23

Funtion points

Category	Count	Weight	Points
External Inputs (EI)	3	4	12
External Outputs (EO)	2	5	10
External Inquiries (EQ)	1	4	4
Internal Logical Files (ILF)	2	10	20
External Interface Files (EIF)	2	7	14
Total function points			60

